

Direct and Inverse Proportion

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

One pen costs 40p.

How much do 5 pens cost?

.....

2 [2 marks]

4 identical books cost £26.

Work out the cost of 7 of these books.

.....

3 [2 marks]

3 workers can build a wall in 12 days.

How long would 6 workers, all working at the same rate, take?

.....

4 [2 marks]

200 g of cheese costs £3.20

Work out the cost of 350 g of the same cheese.

.....

5 [3 marks]

y is directly proportional to x . When $x = 4$, $y = 20$

(a) Work out y when $x = 7$

(b) Work out x when $y = 45$

.....

6

[3 marks]

5 machines fill 800 bottles in an hour.

How many bottles would 8 such machines fill in an hour?

7

[3 marks]

6 people take 10 hours to paint a fence.

How long would 4 people take, working at the same rate?

8

[3 marks]

A recipe for 6 people uses 900 g of pasta.

How much pasta is needed for 10 people?

9

[4 marks]

y is directly proportional to x . When $x = 3$, $y = 21$

(a) Find the value of the constant of proportionality.

(b) Write a formula for y in terms of x .

(c) Work out y when $x = 11$

(d) Write down where the graph of y against x crosses the y -axis.

10

[4 marks]

For each pair of quantities, state whether they are in direct proportion or inverse proportion.

(a) The number of workers and the time to finish a fixed job.

(b) The number of litres of petrol bought and the total cost.

(c) The average speed and the time taken for a fixed journey.

(d) The number of identical tins and their total mass.

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

y is directly proportional to x , and $y = 12$ when $x = 3$
Find a formula for y in terms of x .

.....

2

[2 marks]

y is inversely proportional to x , and $y = 5$ when $x = 4$
Find a formula for y in terms of x .

.....

3

[3 marks]

y is proportional to x^2 . When $x = 3$, $y = 45$
Work out y when $x = 5$

.....

4

[3 marks]

y is proportional to $\sqrt{x}$. When $x = 16$, $y = 12$
Work out y when $x = 25$

.....

5

[4 marks]

y is inversely proportional to x^2 . When $x = 2$, $y = 9$
(a) Find a formula for y in terms of x .
(b) Work out y when $x = 6$
(c) Work out the positive value of x when $y = 4$

.....

6

[4 marks]

The force F newtons between two magnets is inversely proportional to the square of the distance d cm between them.

When $d = 4$, $F = 5$

- (a) Find a formula for F in terms of d .
 - (b) Work out F when $d = 10$
 - (c) Work out d when $F = 20$
-

7

[4 marks]

- (a) y is proportional to x^3 . What happens to y if x is doubled?
 - (b) y is inversely proportional to x^2 . What happens to y if x is trebled?
-

8

[4 marks]

The time T days to build a wall is inversely proportional to the number of workers n .

6 workers take 15 days.

- (a) Find a formula for T in terms of n .
 - (b) How long would 10 workers take?
 - (c) How many workers are needed to finish in 5 days?
-

9

[4 marks]

y is directly proportional to x^2 and inversely proportional to z .

When $x = 2$ and $z = 4$, $y = 3$

Work out y when $x = 4$ and $z = 6$

The distance d metres needed to stop a car is proportional to the square of its speed v mph.

At 30 mph the stopping distance is 45 m.

- (a) Find a formula for d in terms of v .
- (b) Work out the stopping distance at 60 mph.
- (c) Explain the effect on the stopping distance of doubling the speed.
- (d) Find the speed at which the stopping distance is 80 m.